Printed Pages:01 **Sub Code: NME-101**

Paper Id: 233581 **Roll No.** | | | | | | | | | | |

B.TECH
(SEM I) THEORY EXAMINATION 2022-23
BASIC MANUFACTURING PROCESS

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) Define hardness. How is it different with toughness?
(b) Classify steel.
(c) What is core? Where is it used?
(d) What is the difference between up milling and down milling?
(e) Define annealing.
(f) Differentiate between drilling, reaming and boring.
(g) What is the role of inert gases in welding?
(h) Discuss the purpose of heat treatment.
(i) What is the function of runner in casting?
(j) Write the name of any four non-ferrous metals.

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Explain the properties of molding sand. Also explain the different casing defects and their remedies.
(b) Draw neat and clean sketch of lathe. And explain the main components of it.
(c) Explain Gas Welding process. Also explain the flames of the gas welding.
(d) Compare hot working and cold working processes.
(e) Write short notes on the following:
    (i) Electroplating
    (ii) Galvanizing

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Explain the working principal of milling machine with neat and clean sketch.
(b) With the help of schematic sketch, describe the basic working principle and important parts of grinding machine.

4. Attempt any *one* part of the following: 10x1=10

(a) Explain various casting defects and their causes. Also discuss the remedy for prevention of these defects.
(b) With the help of neat sketch explain die casting method. Also write down their limitations.

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5. Attempt any *one* part of the following: 10x1=10

(a) Explain the working of extrusion process with neat and clean sketch.

(b) What is fatigue fracture? Explain S-N curve with neat sketch. Also differentiate between Creep & Fatigue.

6. Attempt any *one* part of the following: 10x1=10

(a) With the help of neat sketch explain wire drawing. Also explain different types of forming defects.

(b) What do you understand by forming? Explain extrusion process and with the help of neat sketch define all important process comes in this process.

7. Attempt any *one* part of the following: 10x1=10

(a) What do you mean by Heat treatment? What are the objectives of heat treatment? Compare carburizing and nitriding processes.

(b) What is plant layout? Explain its different types with neat sketch.

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